

Paul Prae

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PROFESSIONAL SUMMARY

AI and data engineering leader with 13+ years building production systems for healthcare, life sciences, and financial services. Currently managing enterprise healthcare data infrastructure and developing AI agents at Arine, a platform serving 45+ health plans and >30M members. Previously held senior AI roles at Booz Allen Hamilton, Amazon Web Services, and Slalom Consulting, delivering solutions for Fortune 500 clients across HIPAA and FedRAMP environments. Deep technical expertise spanning cloud architecture (AWS, Azure), data platforms (Snowflake, PostgreSQL, dbt), and AI systems (LLMs, multi-agent frameworks, generative AI).

PROFESSIONAL EXPERIENCE

Staff AI DataOps Engineer

Arine | Remote | Sep 2025 – Present

- Managed enterprise data platform on Snowflake and AWS, processing petabytes of healthcare data from hundreds of sources for Arine's medication-optimization platform (Inc. 5000 #5 in AI, 45+ health plans, >30M client members)
- Developed HIPAA-compliant AI coding assistants – a Data Engineering Agent that autonomously generated, deployed, and tested ETL pipelines, and a Dev Environment Agent that eliminated multi-day developer onboarding
- Built observability pipeline for Arine's autonomous pharmacist AI agent, ingesting production behavior data from DynamoDB into Snowflake, transforming with dbt, and visualizing metrics in QuickSight dashboards
- Led AI enablement workshops training up to 100 engineers on AI-first development; designed the AI DataOps Toolkit with governance standards, reference architectures, and change management procedures
- Built high-throughput CDC pipelines from PostgreSQL to Snowflake using serverless, event-driven architectures (Lambda, Step Functions, EventBridge)

Chief AI Architect, Senior Manager

Booz Allen Hamilton | Atlanta, GA | Jul 2024 – Mar 2025

- Led AI and analytics solution development at a Fortune 500 firm (27,000+ employees, \$9B+ revenue), designing AI products harnessing clinical, behavioral, and genomic data for population health and predictive medicine
- Delivered AI solutions across FedRAMP and HIPAA-compliant environments, collaborating with cross-functional teams of engineers, data scientists, and healthcare domain experts
- Guided healthcare and life science clients on responsible AI deployment, addressing algorithmic bias, data privacy, and regulatory compliance

Neuroinformatics Data Platform Architect

TRENDS Center | Atlanta, GA | Jan 2022 – Sep 2023

- Served as tech lead for COINSTAC, an open-source federated analysis platform for neuroimaging and genomics data with differential privacy; co-authored peer-reviewed paper in Frontiers in Neuroinformatics
- Architected AWS cloud infrastructure with GPU-focused compute for deep learning on brain imaging and omics data
- Drove platform growth through open-source contributor recruitment, SBIR grant writing, and industry partnerships

Chief AI Officer, Founder

Hyperbloom | United States | Jan 2020 – Sep 2025

- Grew AI consulting business to \$1.4M ARR from a \$40K investment, delivering cloud-based AI and data solutions for healthcare, life science, and financial services clients
- Architected disaster recovery and data governance solutions for clinical trial software supporting 10,000+ research sites across 45 countries with 5.5M monthly remote monitoring activities

Enterprise AI and ML Solutions Architect

Amazon Web Services | Atlanta, GA | Aug 2018 – May 2021

- Managed portfolio of 10+ enterprise accounts including Cox Enterprises, Cox Automotive, Equifax, NCR, Accenture, and Deloitte, providing prescriptive AI/ML architecture guidance on the AWS platform
- Engaged nationally as a subject matter expert in AI and machine learning, authoring technical white papers, reference architectures, and proof-of-concept solutions that drove production adoption
- Partnered with C-suite executives on technical and business strategy including go-to-market, cost management, and technical recruiting; spoke at public and private AWS events

Senior AI Engineer

NeuroLex Labs | Atlanta, GA | Jan 2018 – May 2020

- Architected automated ML pipelines for healthcare voice computing — collecting, cleaning, training, and deploying models processing text and audio data for neurodegenerative disease prediction
- Drove test-driven development adoption across the engineering team; mentored research fellows in the Tribe program focused on prediction and prevention of neurodegenerative diseases

Advanced Analytics Consultant

Slalom Consulting | Atlanta, GA | Jul 2015 – Jan 2018

- Delivered 7+ client engagements generating \$2M+ in revenue, including a behavioral healthcare application and a statewide demand forecasting system for behavioral health crisis services across Georgia
- Built conversational AI assessment comparing Azure Bot Service to Amazon Lex for a Fortune 100 insurance company; recognized nationally as a subject matter expert in NLP and machine learning

Software Engineer

Red Ventures | Fort Mill, SC | Mar 2014 – Jul 2015

- Engineered full-stack marketing platform serving 16+ Fortune 500 clients (DirecTV, Verizon, AT&T, Liberty Mutual, Progressive) across hundreds of high-traffic consumer-facing sites

Senior Support Engineer

Microsoft | Charlotte, NC | Jul 2012 – Mar 2014

- Delivered advanced technical support to Fortune 100 clients (Chevron, Walmart, Disney, Intel, DoD) for hundreds of high-impact web applications and server farms; MACH Program graduate, nominated for “Entrepreneur of the Year” 2013

EDUCATION

BS, Computer Science (AI Emphasis) & BA, Cognitive Science (AI

Foundation) University of Georgia | 2008 – 2012 Graduated Cum Laude | Dean’s List | HOPE Scholarship

TECHNICAL SKILLS

AI & Machine Learning: Generative AI, Large Language Models (LLMs), Multi-Agent Systems, Natural Language Processing, MLOps, Deep Learning, Prompt Engineering, Predictive Analytics

Cloud & Infrastructure: AWS (Bedrock, SageMaker, ECS/EKS, Lambda, Step Functions, S3, DynamoDB, Glue, QuickSight), Microsoft Azure, Snowflake, Docker, Distributed Systems

Data Engineering: PostgreSQL, dbt, Apache Kafka, ETL/CDC Pipelines, Data Governance, HIPAA/FedRAMP Compliance, Differential Privacy

Programming Languages: Python, SQL, JavaScript, TypeScript, C#, Bash

AI Development Tools: Cursor AI, Claude Code, GitHub Copilot, LangChain, Ollama, n8n, Neo4j, MCP

Leadership & Strategy: Engineering Team Management, AI Strategy & Governance, Executive Coaching, Technical Pre-Sales, Training Delivery

CERTIFICATIONS

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- **AWS Certified Machine Learning Engineer** — Amazon Web Services (Sep 2025)
 - **AWS Certified AI Practitioner** — Amazon Web Services (Sep 2025)
 - **AWS Certified Machine Learning – Specialty** — Amazon Web Services (Jan 2020)
 - **AWS Certified Solutions Architect** — Amazon Web Services (Jul 2018)

PROJECTS

Modular Earth

Not-for-profit open-source initiative building AI agents to help working-class families generate wealth. Active projects include a career strategy assistant and a financial planning assistant built on a local-first, privacy-first stack (Ollama, LangChain, n8n, Neo4j) with zero cloud dependency.

PUBLICATIONS

Enhancing Collaborative Neuroimaging Research: Introducing COINSTAC Vaults for Federated Analysis and Reproducibility

Frontiers in Neuroinformatics (Jun 2023). Co-authored peer-reviewed paper on federated analysis infrastructure enabling privacy-preserving neuroimaging research across institutions without sharing raw data.